

Table 49.3 Cumulative Failure Rates

Maximum Price Rise (%)	Bull Market	Bear Market
5 (breakeven)	684 or 8%	212 or 8%
10	968 or 19%	431 or 23%
15	851 or 28%	344 or 36%
20	749 or 37%	318 or 48%
25	564 or 43%	283 or 58%
30	556 or 50%	218 or 66%
35	465 or 55%	162 or 72%
50	1,032 or 66%	328 or 84%
75	1,109 or 79%	224 or 92%
Over 75	1,848 or 100%	226 or 100%

How do you read the table? Let me share a few examples. In bull markets, 8% of pipes will fail to rise more than 5% after the breakout. The failure rate doubles in the next row and is triple the breakeven rate at 15%. In other words, 28% of patterns will fail to see price rise more than 15%. Bear markets are even worse, as one would expect, for a bullish pattern fighting against a bearish tide.

Use the table to estimate how likely it will be to meet your cost of doing business and any profit margin you want to maintain. For example, say your cost is 5% and you want to make 15% on each trade, for a 20% total.

How many patterns fail to rise more than 20%? In bull markets, 37% will fail and in bear markets, the results are worse, with 48% not exceeding a 20% threshold.

Table 49.4 shows an abbreviated breakout-related statistics table. I don't show throwbacks because they must happen in less than a month (four price bars on the weekly scale, making them scarce), nor gaps, which are rare on the weekly scale.

Breakout direction. By definition, the breakout is upward from a pipe bottom. If price closes below the bottom of the pattern before closing above the top of it, then you don't have a valid pipe bottom.